



INSTITUTE OF AERONAUTICAL ENGINEERING

(Autonomous)

Dundigal, Hyderabad-500043

CSE (ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

QUESTION BANK

Course Title	COMPUTER VISION				
Course Code	ACCD05				
Program	B.Tech				
Semester	VI	CSE (AI & ML)			
Course Type	Elective				
Regulation	BT-23				
Course Structure	Theory			Practical	
	Lecture	Tutorials	Credits	Laboratory	Credits
	3	-	3	-	-
Course Coordinator	Mr. V VIDYASAGAR, Assistant Professor				

COURSE OBJECTIVES:

The students will try to learn:

I	I. The theoretical and practical aspects of computing with images and connect issues from computer vision to human vision.
II	II. The foundation of shape and region analysis and understand the basics of 2D and 3D computer vision.
III	III. The Hough Transform and its applications to detect lines, circles, ellipses. IV. The applications related to computer vision algorithms..

COURSE OUTCOMES:

After successful completion of the course, students should be able to:

CO1	Identify basic concepts, terminology, theories, models and methods in the field of computer vision.	Remember
CO2	Understand fundamental concepts and principles of computer vision, including image filtering, edge detection, and feature extraction.	Understand
CO3	Implement various concepts of shapes and regions for the applications of object detection, recognition, medical image analysis and robotics and automation systems etc.	Apply

CO4	Apply theHoughtransformtechniquetodetect lines,circlesand shapes involved in the image.	Apply
CO5	Understand 3Dgeometry principlestofindthespatial relationshipsbetweenobjectsinascene	Understand
CO6	Develop theabilitytoapply computervisiontechniques to real-worldproblemsandevaluatetheirperformance.	Apply

QUESTIONBANK:

Q.No	QUESTION	Taxonomy	Howdoesthissubsume the level	CO's
MODULE I				
IMAGEPROCESSINGFOUNDATIONS				
PARTA-PROBLEMSOLVINGANDCRITICALTHINKINGQUESTIONS				
1	Why is a comprehensive review of image processing techniques important in the field of computer vision and image analysis? How does a thorough understanding of these techniques benefit various applications, from medical imaging to content creation? a) How have image processing techniques evolved over time, and what are the key factors driving their advancements? b) What are the historical developments and milestones in the field of image processing? c) How is image enhancement used in medical imaging to improve the visibility of critical structures and anomalies?	Understand	The learner has to recall the concept of image processing techniques. Identify keypoints and descriptors in an image, used for tasks like object recognition and matching..	CO1

2	Analyze the challenges and limitations of classical filtering operations, particularly in cases involving complex scenes, mixed noise sources, or real-time processing requirements. How do these factors impact the choice and effectiveness of filtering techniques? a) What strategies can be employed to handle mixed noise sources in images, such as salt-and-pepper noise combined with Gaussian noise? b) How do real-time processing constraints influence the selection of appropriate filtering methods for applications like video analysis? c) What are some applications of edge detection filters like the Sobel filter?	Understand	The learner has to recite Classical filtering operations in image processing utilize convolution make use of it.	CO1
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3	How Thresholding methods segment images into binary or multi-level representations based on pixel intensity. They help isolate objects of interest and simplify further analysis, with applications in object detection and image segmentation. a) How can you determine an appropriate threshold value for a given image? b) How does global thresholding differ from local thresholding? c) Explain how adaptive thresholding works and its advantages.	Understand	The learner has to recall the Thresholding methods segment images into binary or multi-level and demonstrate the objects of interest and simplify further analysis,	CO1
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4	How Edge detection algorithms identify abrupt intensity changes in images, highlighting object boundaries. Popular methods include the Canny edge detector, which applies Gaussian smoothing and gradient analysis, and the Laplacian of Gaussian (LoG) operator. a) Why is edge detection crucial in image processing? b) Can you name some real-world applications where edge detection is vital? c) How does the Laplacian of Gaussian operator enhance edges?	Understand	The learner has to recall the Edge detection algorithms identify abrupt intensity changes in images, highlighting object boundaries. Popular methods include the Canny edge detector .	CO1
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5	How Corner and interest point detection identifies distinctive image features used in tasks like image stitching and object tracking. Algorithms, such as the Harris corner detector, locate points with significant variations in local intensity.”. a) How does the Harris corner detector work to find corners in an image? b) In what applications are corner and interest point detection commonly used? c) In what applications are corner and interest point detection commonly used?	Apply	The learner has to recall the Corner and interest point detection identifies distinctive image features used in tasks like image stitching and object tracking. Algorithms, .	CO1
6	Compare Euclidean distance and Hausdorff distance as metrics for shape comparison. Discuss cases where one distance measure might be more sensitive to shape variations than the other.	Understand	The learner has to recall the Euclidean distance and Hausdorff distance as metrics for shape comparison	CO2
7	How do different corner and interest point detection algorithms work?	Understand	The learner has to recall the Compare and contrast the principles behind popular algorithms like Harris, Shi-Tomasi, and Scale-Invariant Feature Transform (SIFT)	CO2

8	Can corner and interest point detection be combined with other computer vision techniques?	Understand	The learner has to recall the concept of these points are used in conjunction with image segmentation, object tracking, and 3D reconstruction.	CO2
9	How have recent advancements in deep learning affected corner and interest point detection??	Understand	The learner has to recall the Explore the integration of deep neural networks with traditional feature detection methods.	CO1
10	How do human visual perception and interpretation of texture compare to computer-based texture analysis?	Understand	The learner has to recall the concept human visual perception and interpretation of texture.	CO2
PART-B LONG ANSWER QUESTIONS				
1	Summarize the image processing techniques?	Understand	The learner has to recall and methods used to manipulate and analyze images for various applications. extend them with neat diagrams.	CO1
2	How do filters contribute to image processing?	Understand	The learner has to recall Filters enhance images by applying convolutional kernels identifies cryptanalysis highlight specific features or remove noise.	CO1
3	What is the primary purpose of object detection algorithms?	Understand	The learner has to recall Object detection algorithms locate and classify outline objects within images or video streams.	CO1
4	Summarize Histogram Equalization and Morphological Operations	Understand	The learner has to recall and Morphological operations involve the modification of the shape identify structure of objects in an image	CO1

5	When is adaptive thresholding preferred over global thresholding?	Apply	The learner has to recall the concept of Adaptive thresholding and relate Adaptive thresholding is preferred when an image has uneven illumination or when different regions have varying contrast	CO1
6	What are edge detection techniques, What are the key steps involved in edge detection?	Understand	The learner has to identify the concept to identify the boundaries or edges of objects within an image.	CO1
7	State the details about the following: a) Linear Filtering b) Non-linear Filtering c) Frequency Domain Filtering d) Edge Detection Filters e) Morphological Filters f) substitution techniques	Understand	The learner has to recall the concept of security attacks, security services, security mechanism, plain text, ciphertext, substitution techniques, and transposition techniques.	CO1
8	What are the challenges and limitations of edge detection techniques? a) Complex Scenes b) Computational Complexity	Apply	The learner has to recall the enabling machines to perceive and interpret visual information in images and videos, extend the transposition techniques. Make use the wide range of practical applications across various industries.	CO1
9	Summarize some edge detection algorithms? a) Laplacian of Gaussian b) Robert's Cross Operator	Understand	The learner has to recall the concepts of edge detection algorithms, mechanisms, and infer about them.	CO1

10	What are some common corner detection algorithms?	Understand	The learner has recall the concept of corner detection algorithms. extend This algorithm measures the intensity change in all directions and calculates a corner response based on the eigenvalues of the structure tensor.	CO1
11	Summarize the key characteristics of corner points and interest points?	Understand	The learner Corner points are located at regions where there are significant changes in multiple directions,	CO2
12	What is mathematical morphology in the context of computer vision, and how does it contribute to image processing?	Apply	The learner has to relate Mathematical morphology is a framework in image processing and computer vision that focuses on the analysis and manipulation of the shape and structure of objects within images.	CO2
13	What are the fundamental operations in mathematical morphology, and how do they work?	Apply	The learner has to list the concept of cipher text by make use of hill cipher and utilise it to calculate the encryption for the given plaintext.	CO2
14	What are some applications of texture analysis in computer vision?	Apply	The learner has to list the concept of texture analysis make use of Distinguishing objects based on their surface textures.	CO2
15	How is texture information useful in medical image analysis?	Apply	The learner has to list the concept of texture analysis helps identify regions of interest make use of In medical imaging	CO2

16	How does texture segmentation work?	Apply	The learner has to relate the concept of by texture segmentation work using image into regions based on similar textural properties. apply texture analysis techniques to identify regions with consistent textures and then grouping them together.	CO2
17	How are thresholding techniques used in object detection and image analysis?	Understand	The learner has to list the in various object detection and image analysis tasks and extend its applications.	CO2
18	Structuring Elements and Shape Analysis	Understand	The learner has to list the concepts How does its shape and size impact morphological operations	CO2
19	Summarize the Morphological Skeletonization and Pruning	Understand	The learner has to recall the concept of the importance of pruning in morphological skeletonization.	CO2
20	Explain the Advanced Topics in Mathematical Morphology	Understand	The learner has to recall the concept of it extend the binary operations to gray-scale images?	CO2
PART-C SHORT ANSWER QUESTIONS				
1	Define image processing?	Remember	The learner has to define the concept of image processing	CO1
2	Define classical filtering operation?	Remember	The learner has to recall definition of classical filtering operation .	CO1
3	Define thresholding in image processing?	Understand	The learner has to Recall thresholding in image processing and identify intensity values.	CO1
4	Define an edge detection technique.?	understand	The learner has to relate edges in images by calculating gradient magnitudes identify its applications.	CO1

5	What does corner detection aim to find?	Understand	The learner has to recall the definition of corner detection and extend the different examples.	CO1
6	Name an interest point detection algorithm?	Understand	The learner has to recall the concept of interest point detection algorithm outline key principles of interest point detection algorithm.	CO1
7	What is mathematical morphology in image processing?	Understand	The learner has to recall the concept of mathematical morphology.	CO1
8	Explain how erosion works in mathematical morphology?	Understand	The learner has to recall the concept of mathematical morphology extend its applications	CO1
9	What is the purpose of texture analysis in image processing.	Understand	The learner has to recall texture analysis.	CO1
10	Which technique is used for texture analysis??	Understand	The learner has to Recall texture analysis and explain its applications	CO1
11	What is an advantage of classical filtering operations??	Understand	The learner has to Recall the concept of . classical filtering operations	CO2
12	What is the drawback of simple thresholding?	Understand	The learner has to recall the simple thresholding?	CO2
13	How does SIFT handle image scaling and rotation?	Remember	The learner has to recall the concept of SIFT handle image scaling and rotation.	CO2
14	What application uses interest point detection?	Remember	The learner has to relate the concept cryptanalysis.	CO2
15	Why are erosion and dilation used together in image processing.	Remember	The learner has to recall erosion and dilation.	CO2
16	Give an example of texture analysis application?	Remember	The learner has to recall the concept security approaches.	CO2
17	What is the significance of the GLCM in texture analysis?	Understand	The learner has to recall the concept of GLCM in texture analysis and Extend its applications.	CO1

18	How can image processing improve medical diagnoses?	Understand	The learner has to recall the concept of image processing	CO2
hline 19	How does the Sobel operator detect edges?	Understand	The learner has to recall the concept of Sobel operator detected edges and Extend its applications.	CO2
20	What is a common challenge in image processing?	Remember	The learner has to list various image processing.	CO2

MODULE II

SHAPES AND REGIONS

PART-A PROBLEM SOLVING AND CRITICAL THINKING QUESTIONS

1	Binary shape analysis involves the study of objects in binary images. It encompasses methods to describe, compare, and classify shapes based on their geometric properties, which find applications in pattern recognition and image understanding. a) How do binary shape analysis methods handle objects with irregular shapes? b) Can you provide an example of a problem where binary shape analysis is crucial? c) How are shape descriptors calculated for binary objects?	Understand	The learner has to recall the concept of Binary shape analysis involves the study of objects in binary images.	CO3
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2	How Connectedness analysis focuses on identifying distinct objects in an image and labeling them. It is used to count and differentiate individual objects or regions, a fundamental step in many image analysis tasks. a) How does object labeling help in counting and distinguishing objects in an image? b) What are some common algorithms for connected component labeling? c) Can you explain how connectedness analysis handles overlapping objects?	Understand	The learner has to recall the concept of Connectedness analysis focuses on identifying distinct objects in an image and labeling them	CO3
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3	Size filtering is a technique used to retain or remove objects in an image based on their size. It's commonly used in image processing for noise reduction, feature extraction, and object separation.? a) How does size filtering distinguish between small and large objects? b) Can you provide examples of applications where size filtering is essential? c) How does size filtering affect the computational complexity of an image processing pipeline?	Apply	The learner has to recall Size filtering is a technique used to retain or remove objects in an image based on their size.	CO3
4	Skeletonization and thinning algorithms extract the skeleton or medial axis of objects, useful for shape analysis and representation.? a) How do skeletonization and thinning algorithms work to extract object skeletons? b) In what scenarios is skeletonization particularly useful? c) Can you explain the concept of the medial axis of an object?	Understand	The learner has to recall the concept of Skeletonization and thinning algorithms extract the skeleton or medial axis of objects.	CO3

5	Deformable shape analysis involves modeling and tracking objects that can change shape over time. Active contours, also known as snakes, use energy minimization to detect object boundaries, and shape models represent object variability. a) Why is deformable shape analysis important in computer vision? b) How do active contours adapt to object boundaries in images? c) Can you explain the concept of energy minimization in active contours?	Apply	The learner has to the relate Deformable shape analysis involves modeling and tracking objects that can change shape over time. Active contours	CO3
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6	How Centroidal profiles represent objects' shapes based on their centroids, aiding in shape characterization. Handling occlusion involves addressing situations where objects are partially hidden, and boundary length measures quantify object perimeters. a) What is the role of centroidal profiles in shape representation? b) How do image processing techniques handle occluded objects? c) Can you provide examples of centroidal profile applications?	Understand	The learner has to recall the concept of Centroidal profiles represent objects' shapes based on their centroids, aiding in shape characterization. Handling occlusion involves addressing situations where objects are partially hidden, and boundary length measures quantify object perimeters..	CO3
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7	How Boundary descriptors are numerical representations of object shapes. Chain codes encode object boundaries, and Fourier descriptors capture shape information in the frequency domain a) How do boundary descriptors aid in shape representation? b) How do Fourier descriptors differ from other shape descriptors? c) What are the advantages and limitations of each of these methods in shape analysis?	Understand	The learner has to recall Boundary descriptors are numerical representations of object shapes. Chain codes encode object boundaries, and Fourier descriptors capture shape information in the frequency domain..	CO3
8	. What information do region descriptors provide about objects?	understand	The learner has to recall the concept of Region descriptors characterize objects based on their pixel values within the object..	CO3
9	What challenges can arise when using region descriptors with complex objects in images?.	Apply	The learner has to recall the concept of Region descriptors characterize objects based on their pixel values within the object. Moments are statistical measures of object shapes used for feature extraction and pattern recognition..	CO3
10	What types of information do moments convey about object shapes?	Understand	The learner has to recall concept of Moments are statistical measures of object shapes used for feature extraction and pattern recognition.	CO3
PART-B LONG ANSWER QUESTIONS				

1	summerize the Shape Descriptors and Features	Understand	The learner has to recall simple shape descriptors Compute they are computed.	CO3
2	Explain the Applications connectedness.	Apply	The learner has to recall connected components analysis be used for object detection and labeling extend algorithmic explanation	CO3
3	explain the Object Counting Strategies	Understand	The learner has to recall concept of Object Counting Strategies compare global object counting strategies.	CO3
4	Describe the process of size filtering using morphological operations.	Understand	The learner has to relate the concept process of size filtering and summerize involve the use of dilation and erosion .	CO3
5	What are distance functions in computer vision, and why are they important for various tasks?	Understand	The learner has to recall the distance functions important for various tasks?	CO3
6	What is thinning in the context of image processing and computer vision? How is thinning different from other image simplification techniques?	Understand	The learner has to recall concept of thinning in the context of image processing	CO3
7	Explain the concept of point-based deformable registration. How is it applied in medical imaging?	Understand	The learner has to recall the concept of point-based deformable registration. and outline its principles.	CO2
8	Compare and contrast contour-based tracking with region-based tracking techniques.	Understand	The learner has to relate the concept of contour-based tracking with region-based tracking techniques.	CO3
9	What are active contours in vision, and how do they work? Provide examples of applications where active contours are used.?	Understand	The learner has to relate the concept of active contours and outline applications where active contours are used.	CO3

10	What are shape models in computer vision, and how do they play a role in various applications? Provide examples of shape modeling techniques and their significance.	Understand	The learner has to relate the concept shape models techniques and their significance	CO3
11	What are centroidal profiles in computer vision, and how are they used in shape analysis and recognition? Provide examples of applications where centroidal profiles play a significant role.	Understand	The learner has to recall the concept centroidal profiles extend applications centroidal profiles play a significant role	CO3
12	What are some strategies for handling occlusion, How can object detection and tracking algorithms be made robust to occlusion?	Understand	The learner has to recall the handling occlusion extend its advantages and limitations.	CO3
13	How do boundary length measures handle scale, noise, and occlusion challenges? What are their limitations in complex scenarios?	Understand	The learner has to recall the concept of boundary length measures handle scale and limitations in complex scenarios.	CO3
14	What are chain codes in the context of computer vision? Provide a detailed explanation of their purpose and how they work.?	Understand	The learner has to list the concept chain codes.	CO3
15	How are chain codes used in practical applications of computer vision? Provide examples of real-world applications where chain codes play a significant role.	Understand	The learner has to recall where compact and efficient representation of object boundaries is important. outline efficient storage of shape information and quantitative analysis of medical images.	CO3

16	How are Fourier descriptors applied in real-world applications of computer vision? Provide examples of practical scenarios where Fourier descriptors are utilized.	Understand	The learner has to recall Fourier descriptors applied in real-world Extend its applications	CO3
17	What are the advantages and limitations of using Fourier descriptors for shape analysis in computer vision? Provide examples to illustrate your points.	Understand	The learner has to recall the concept Fourier descriptors for shape analysis .	CO3
18	What are region descriptors in the context of vision? Explain their significance and provide a detailed overview of how they work.	Understand	The learner has to relate the concept of region descriptors and outline capture various visual properties and attributes of the content within the region.	CO3
19	How are region descriptors applied in practical applications of computer vision? Provide examples of real-world scenarios where region descriptors play a significant role.	Understand	The learner has to recall the concept of region descriptor extend its uses.	CO3
20	What are moments in the context of computer vision? Explain their significance and provide a detailed overview of the different types of moments and their applications.	Understand	The learner has to recall moments summarize its applications	CO3
PART-C-SHORT ANSWER QUESTIONS				
1	What is binary shape analysis?	Understand	The learner has to relate the concept binary shape analysis and outline them with examples.	CO3
2	How is connectedness defined in binary shape analysis?	Understand	The learner has to recall connectedness defined in binary shape analysis .	CO3

3	What is object labeling and counting in binary shape analysis?	Understand	The learner has to recall object labeling and counting .	CO3
4	How does size filtering work in binary shape analysis?	Understand	The learner has to recall the concept of size filtering working.	CO3
5	What are distance functions used for in binary shape analysis??	Understand	The learner has to recall the concept of distance functions and extend Applications on it.	CO3
6	What is the difference between skeletonization and thinning?	understand	The learner has to recall skeletonization and thinning and identify its importance.	CO3
7	What is deformable shape analysis?	Understand	The learner has to recall the deformable shape analysis and extend its uses.	CO3
8	How do boundary tracking procedures work?	Understand	The learner has to Recall boundary tracking procedures .	CO3
9	What are active contours used for in image processing?	Understand	The learner has to recall the concepts of active contours used for in image processing and outline its advantages.	CO3
10	How do shape models aid in shape recognition?	Understand	The learner has to recall differential and linear cryptanalysis.	CO3
11	How do shape models aid in shape recognition?	Understand	The learner has to recall shape recognition and outline its applications.	CO3
12	How does binary shape analysis handle occlusion??	Understand	The learner has to recall the concept handle occlusion and demonstrates it with example.	CO2
13	What are boundary length measures used for in binary shape analysis?	Understand	The learner has to recall the concept boundary length measures and extend its procedure.	CO2

14	Illustrate the procedure for RSA algorithm?	Understand	The learner has to relate the concept of RSA algorithm and demonstrates the procedure for RSA algorithm.	CO2
15	Illustrate RSA, Diffie-Hellman, ECC Key Distribution Algorithm?	Understand	The learner has to recall the concept of RSA Diffie-Hellman, ECC Key Distribution Algorithm and illustrate them.	CO2
16	List key distribution Asymmetric key Ciphers?	Understand	The learner has to label the key distribution asymmetric key ciphers.	CO2
17	Demonstrate link and end-to-end encryption?	Understand	The learner has to recall the concept of link and end-to-end encryption.	CO2
18	Recall session key and master key?	Apply	The learner has to relate the concept of session key and master key.	CO2
19	Recognise the design criteria of block cipher?	Apply	The learner has to recall the design criteria of block cipher.	CO2
20	Write placement of encryption function?	Remember	The learner has to outline the placement of encryption function.	CO2
MODULE III				
HOUGH TRANSFORM				
PART A- PROBLEM SOLVING AND CRITICAL THINKING QUESTIONS				

1	How Line detection in images involves finding straight lines. The Hough Transform (HT) is a technique used to detect lines by transforming image space into parameters space, simplifying the detection process. a) How does the Hough Transform simplify the detection of lines? b) What are the parameters that the Hough Transform uses for line detection? c) How does line fitting work in the context of the Hough Transform?	Apply	The learner has to recall the concept of Line detection in images involves finding straight lines. The Hough Transform (HT) is a technique used to detect lines by transforming image space into parameters space, simplifying the detection process.	CO4
2	RANSAC (Random Sample Consensus) is a robust algorithm used with the Hough Transform for detecting straight lines in noisy images. It iteratively fits lines to subsets of data points to find the best fit. a) How does RANSAC handle noisy data when detecting lines? b) Can you explain the iterative process of RANSAC for straight line detection? c) What criteria are used to determine the "best fit" line in RANSAC?	Understand	The learner has to recall the concept RANSAC (Random Sample Consensus) is a robust algorithm used with the Hough Transform for detecting straight lines in noisy images	CO4

3	How Hough Transform can be adapted for detecting circular objects in images. It identifies circles by transforming the image space into parameter space, allowing accurate center location and radius estimation. a) How does the Hough Transform differ when applied to circle detection? b) How does the Hough Transform provide accurate center location of circles? c) What challenges can arise when using the Hough Transform for circle detection?	Remember	The learner has to recall The Hough Transform can be adapted for detecting circular objects in images. It identifies circles by transforming the image space into parameter space, allowing accurate center location and radius estimation..	CO4
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4	How the Generalized Hough Transform (GHT) is an extension of the Hough Transform used for detecting arbitrary shapes and objects. It performs spatial matched filtering to locate objects in an image. a) How does the Generalized Hough Transform differ from the standard Hough Transform? b) How does GHT facilitate feature collation in image analysis? c) Can you provide an example of a practical application where GHT is used for object location?	Understand	The learner has to recall the concept. The Generalized Hough Transform (GHT) is an extension of the Hough Transform used for detecting arbitrary shapes and objects. It performs spatial matched filtering to locate objects in an image.	CO4
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5	In the context of human iris recognition, the Hough Transform and Generalized Hough Transform can be used for iris location and hole detection. GHT is employed for ellipse detection, aiding in accurate feature collation. a) How does the Hough Transform assist in locating the human iris in iris recognition systems? b) How is the Generalized Hough Transform utilized for ellipse detection in the iris? c) Are there speed-related issues when applying these techniques to iris recognition systems, and how can they be addressed?	Understand	The learner has to relate the concept of human iris recognition, the Hough Transform and Generalized Hough Transform can be used for iris location and hole detection. GHT is employed for ellipse detection, aiding in accurate feature collation.	CO4
6	How does line fitting help in improving the accuracy of detected lines?	Understand	The learner has to recall concept of line fitting Summarize by adjusting the detected line's parameters to best match the actual data points in the image.	CO4
7	What is the RANSAC algorithm and how is it related to straight line detection in the Hough Transform?	Understand	The learner has to recall the concept of message authentication and identify types of attacks are addressed by message authentication.	CO4

8	How does the Generalized Hough Transform differ from the standard Hough Transform, and why is it necessary for ellipse detection?	Understand	The learner has to recall the concept of Generalized Hough Transform differ from the standard Hough Transform, and why is it necessary for ellipse detection?.	CO4
9	What are some real-world applications where ellipse detection using GHT is particularly useful? Can you think of scenarios where other shape detection methods might be more appropriate?.	Understand	The learner has to recall the concept of real-world applications where ellipse detection using GHT is particularly useful.	CO4
10	In the context of real-time computer vision applications, what strategies can be employed to accelerate GHT-based ellipse detection and make it suitable for high-speed processing?	Understand	The learner has to recall to accelerate GHT-based ellipse detection and make it suitable for high-speed processing.	CO4
PART-B LONG ANSWER QUESTIONS				
1	Explain in detail how the Hough Transform can be utilized for accurate line localization. Provide a step-by-step explanation of the process, including the challenges faced, methods to enhance accuracy, and practical applications of this approach?	Understand	The learner has to recall the concept of line localization and extend accurately detecting and localizing lines within an image, even when they are incomplete, curved, or obscured by noise .	CO4
2	How does the Hough Transform work for line detection?	Understand	The learner has to recall the Hough Transform work for line detection	CO4
3	Are there any strategies to improve the accuracy of line localization?	Understand	The learner has to recall techniques to achieve sub-pixel accuracy when determining the line endpoints. outline with an example.	CO4

4	What are the steps involved in performing least squares line fitting?	Understand	The learner has to recall the concept and the concept. Collect the edge points or points corresponding to the detected line from the original image. calculate the slope of the line using the formula: $\text{slope} = ((x - x_{mean}) * (y - y_{mean})) / ((x - x_{mean})^2)$	CO4
5	Are there any limitations to line fitting in the Hough Transform??	Understand	The learner has to recall line fitting and extend . Is line fitting computationally expensive?	CO4
6	How does line fitting help in improving the accuracy of detected lines?	Understand	The learner has recall the concept of line fitting summarize by adjusting the detected line's parameters to best match the actual data points in the image.	CO4
7	What is the RANSAC algorithm and how is it related to straight line detection in the Hough Transform?	Apply	The learner has to Recall RANSAC algorithm.	CO4
8	What are the advantages of using RANSAC for straight line detection over traditional methods?	Understand	The learner has to recall the concept of RANSAC for straight line detection.	CO4
9	How does the Circular Hough Transform handle circle parameter space discretization?	Understand	The learner has to recall the concept of Circular Hough Transform handle	CO4
10	How can the accurate center location of an object in an image be determined?	Understand	The learner has to recall the concept of accurate center location.	CO4
11	How does noise affect the accuracy of center location using image moments?	Understand	The learner has to recall concept of center location and techniques . Gaussian smoothing can be applied before calculating moments.	CO4

12	What is integral image and how can it speed up the Hough Transform?	Understand	The learner has to relate speed problems demonstrates speed-related problems include the need for excessive memory usage, long processing times,	CO4
13	Case Study Human Iris Location Using Image Processing?	Understand	The learner has to recall the concept of Human Iris Location Using Image Processing	CO4
14	case study of hole detection with neat diagram.	Apply	The learner has to recall the concept of hole detection extend challenges tasks including person identification, eye disease diagnosis, and human-computer interaction.	CO4
15	Case study of generalized hough transform?	Understand	The learner has to recall generalized hough transform and extend them with neat diagram.	CO4
16	Case study of spatial matched filtering?	Understand	The learner has to recall concept of spatial matched filtering? extend with neat diagram.	CO4
17	case study of GHT for ellipse detection?	Understand	The learner has to recall the concept GHT for ellipse detection extend with neat diagram.	CO4
18	Case study of object location ?	Understand	The learner has to recall the concept of object location and outline with neat diagram.	CO4
19	Case study of GHT for feature collation?	Understand	The learner has to recall the concept of GHT for feature collation extend with neat diagram.	CO4
20	Explained detection in image processing?	Understand	The learner has to recall the concept of line detection in image processing and	CO4
PART-C SHORT ANSWER QUESTIONS				

1	What is line detection in image processing?	Understand	The learner has to recall the concept of line detection .	CO4
2	Explain the Hough Transform (HT) for line detection?	Remember	The learner has to recall concept Hough Transform (HT) for line detection .	CO4
3	How does the foot-of-normal method work in line detection?	Remember	The learner has to recall the concept foot-of-normal method work in line detection .	CO4
4	What is line localization in line detection?	Remember	The learner has to recall the concept line localization.	CO4
5	What is line fitting in image processing?	Remember	The learner has to recall line fitting in image processing .	CO4
6	How does RANSAC work for straight line detection?	Remember	The learner has to recall RANSAC work for straight line detection .	CO4
7	Explain HT-based circular object detection?	Understand	The learner has to recall the concept of HT-based circular object detection Extend its applications .	CO4
8	How is accurate center location achieved in circle detection ?	Remember	The learner has to Recall the center location achieved in circle detection .	CO4
9	What is the speed problem in circle detection using HT? ?	Understand	The learner has to recall the concept circle detection using HT	CO4
10	How does ellipse detection differ from circle detection?	Remember	The learner has to recall ellipse detection differ from circle detection.	CO4
11	Describe a case study Human Iris location?	Understand	The learner has to recall the concept Human Iris location and outline its steps .	CO4
12	What is hole detection in image processing?	Remember	The learner has to recall hole detection in image processing .	CO4

13	How does the (GHT) differ from the standard HT?	Understand	The learner Generalized Hough Transform has to recall the concept of Generalized Hough Transform and extend its applications.	CO4
14	What is spatial matched filtering in GHT?	Remember	The learner has to recall the concept of spatial matched filtering in GHT and Extend its applications.	CO4
15	How is GHT used for ellipse detection?	Understand	The learner has to recall the concept of ellipse detection extend its services.	CO4
16	How does GHT aid in object location?	Remember	The learner has to recall the concept of GHT aid in object location .	CO4
17	What is feature collation in GHT?	Understand	The learner has to recall the concept of feature collation in GHT and extend its applications.	CO4
18	How does RANSAC improve line detection accuracy?	Remember	The learner has to recall the concept of RANSAC improve line detection accuracy.	CO4
19	What are the benefits of using GHT for shape detection?	Remember	The learner has recall the concept of using GHT for shape detection and Extend its applications.	CO4
20	Describe an application of GHT in real-world scenarios.?	Understand	The learner has to recall the concept application of GHT.	CO4

MODULE IV

3D VISION AND MOTION

PART A- PROBLEM SOLVING AND CRITICAL THINKING QUESTIONS

1	Explain how each of the following equations or expressions can be used for detecting and estimating visual motion in a spatio-temporal image sequence $I(x,y,t)$. Include in your answer the name used to describe each of these general classes of motion extraction models? a) $I(x,y,t) = \nabla \cdot \mathbf{v} \rightarrow I(x,y,t)$ b) (iv) $F(x, y, t) = e^{i(xv_x + yv_y + t)} F(x, y, t)$	Apply	The learner has to recall the concept of detecting and estimating visual motion in a spatio-temporal image sequence	CO5
hline 2	What are the fundamental challenges in 3D vision, and how do various methods address these challenges? a) How do stereo vision and structured light scanning tackle the problem of depth perception? b) What are the limitations of time-of-flight (ToF) sensors in 3D vision, and how can they be mitigated?	Understand	The learner has to recall fundamental challenges in 3D vision.	CO5

3	Explore the role of machine learning and deep learning in enhancing 3D vision methods. a) What hardware advancements are crucial for enabling real-time 3D vision processing? b) How do the computational demands of 3D vision impact the choice of algorithms and hardware platforms?	Apply	The learner has to recall the machine learning and deep learning in enhancing 3D vision methods.	CO5
4	Discuss the mathematical and computational principles behind "Shape from Texture" techniques and the challenges involved in estimating 3D shape from 2D texture information. a) How do the statistical properties of textures, such as orientation, frequency, and regularity, affect the accuracy of shape estimation? b) What role does texture mapping play in improving the quality of 3D reconstructions from texture?	Apply	The learner has to recall the concept mathematical and computational principles behind "Shape from Texture"	CO5

5	Explore real-world applications of "Shape from Texture" in computer vision, robotics, and beyond. What are some specific use cases where this technique is particularly advantageous? a) How is "Shape from Texture" applied in object recognition, surface inspection, or scene reconstruction? b) What are the limitations and challenges when implementing "Shape from Texture" in practical applications?	Understand	The learner has recall the concept applications of "Shape from Texture" in computer vision, robotics, and beyond. What are some specific use cases where this technique is particularly advantageous .	CO5
6	Discuss the various technologies and devices used in active range finding, including LIDAR (Light Detection and Ranging), structured light, and laser scanning. What are their strengths and limitations? a) How do different active range finding techniques vary in terms of range, precision, and resolution? b) What are the cost considerations associated with various active range finding devices?	Apply	The learner has to recall the concept various technologies and devices used in active range finding, including LIDAR (Light Detection and Ranging), structured light, and laser scanning.	CO5

7	Discuss the various methods and techniques used in 3D object recognition, including point cloud processing, 3D feature extraction, and deep learning approaches. What are their strengths and limitations? a) How do traditional 3D feature-based methods compare to deep learning methods in terms of accuracy and robustness? b) What are the computational challenges associated with processing large 3D point clouds?	Understand	The learner has to recall the concept various methods and techniques used in 3D object recognition, including point cloud processing, 3D feature extraction, and deep learning approaches. What are their strengths and limitations .	CO5
8	Examine the real-world applications of 3D object recognition across different domains, such as robotics, augmented reality, medical imaging, and industrial automation. How does the recognition process vary in these contexts? a) How is 3D object recognition used in robotic grasping and manipulation tasks? b) What role does 3D object recognition play in medical diagnostics and surgical planning?	Apply	The learner has to recall the concept of real-world applications of 3D object recognition across different domains, such as robotics, augmented reality, medical imaging,	CO5

9	Discuss the significance of 3D reconstruction in various fields, including computer vision, archaeology, architecture, and medical imaging. How does 3D reconstruction benefit these domains? a) How is 3D reconstruction used in cultural heritage preservation and archaeological site documentation? b) What are the clinical applications of 3D reconstruction in medicine, such as patient-specific surgical planning?	Apply	The learner has to recall the concept of 3D reconstruction in various fields, including computer vision, archaeology, architecture, and medical imaging..	CO5
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10	Discuss the real-world applications and significance of translational alignment in computer vision, such as image registration, object tracking, and image stitching. How does alignment enhance the quality and interpretability of visual data? a) How is translational alignment used in panoramic image stitching to create seamless, wide-angle images? b) What role does translational alignment play in medical image registration for accurate diagnosis and treatment planning?	Understand	The learner has to recall the concept the real-world applications and significance of translational alignment in computer vision, such as image registration, object tracking, and image stitching. How does alignment enhance the quality .	CO5
PART-B LONG ANSWER QUESTIONS				
1	How do Time-of-Flight (ToF) cameras contribute to 3D vision?	Understand	The learner has to recall the concept methods for 3d vision and illustrates its specification.	CO5
2	What challenges are associated with achieving accurate 3D vision??	Understand	The learner has to recall the methods for 3d vision outline the appropriate method for a specific application depends on the desired level of accuracy and the constraints of the environment.	CO5
3	How does the choice of method depend on the specific application of 3D vision?	Understand	The learner has to recall the concept the methods for 3d vision demonstrates real time processing demands.	CO5

4	What are projection schemes in 3D vision, and why are they essential for understanding the 3D world?	Understand	The learner has to recall the concept projection schemes and illustrates its applications .	CO5
5	How does "Photometric Stereo" work, and what kind of information does it provide?	Understand	The learner has to recall concept of Photometric Stereo extends providing information about the object's shape and orientation.	CO5
6	What are the assumptions and limitations of the "Shape from Texture" technique?	Understand	The learner has to recall the concept "Shape from Texture"	CO5
7	What are the advantages of using "Active Range Finding" over passive depth sensing methods?	Understand	The learner has to recall the concept of Active Range Finding .	CO5
8	How do different surface representations capture the shape and geometry of objects?	Understand	The learner has to recall the concept of surface representations identify Volumetric Representations	CO5
9	What role do point-based representations play in non-rigid motion analysis?	Understand	The learner has to recall the concept point based representation and outline their deformations.	CO5
10	How do volumetric representations relate to motion analysis in 3D vision?	Understand	The learner has recall the concept of volumetric representations and identify its specifications.	CO5
11	What challenges exist in 3D object recognition, particularly in dynamic environments or involving moving objects?	Remember	The learner has to recall 3D Object recognition.	CO5
12	What is 3D reconstruction in the context of 3D vision, and why is it important?	Understand	The learner has to recall the concept of 3D reconstruction approaches common approaches used for 3D reconstruction.	CO5
13	What is motion in the context of 3D vision, and why is it important?	Understand	The learner has to recall motion in the context of 3D vision.	CO5

14	What is triangulation in the context of 3D vision, and why is it important?	Understand	The learner has to recall the concept of triangulation in 3D vision and extends the importance of triangulation	CO5
15	How is bundle adjustment used for camera calibration in 3D vision??	Understand	The learner has to recall the concept of adjusting intrinsic and extrinsic camera parameters along with 3D point positions.	CO5
16	What is translational alignment in the context of 3D vision, and why is it important?	Understand	The learner has to recall the concept of translational alignment in 3D vision and extends its importance.	CO5
17	How is parametric motion modeled and represented mathematically?	Understand	The learner has to recall the concept of parametric motion	CO4
18	What types of splines are commonly used for spline-based motion in 3D vision?	Understand	The learner has to recall the concept of spline-based motion and identify the spline types	CO5
19	How does optical flow relate to motion analysis and tracking in 3D vision?	Apply	The learner has to recall the concept of optical flow in 3D vision	CO5
20	How is layered motion used to represent complex scenes in 3D vision?	Understand	The learner has to recall the concept of layered motion identify its applications in real world	CO5
PART-C SHORT ANSWER QUESTIONS				
1	What are methods for 3D vision?	Remember	The learner has to recall the concept of methods for 3D vision	CO5
2	How do projection schemes work in 3D vision?	Understand	The learner has to recall the concept of projection schemes work and identify the applications.	CO5
3	What is shape from shading in 3D vision?	Understand	The learner has to recall the concept of shading in 3D vision .	CO5
4	How does photometric stereo aid in 3D vision?	Remember	The learner has to recall the photometric stereo.	CO5
5	What is shape from texture in 3D vision?	Understand	The learner has to recall the concept of texture in 3D vision and identify its uses .	CO5

6	Explain shape from focus in 3D vision?	Understand	The learner has to recall focus in 3D vision outline its applications	CO5
7	What is active range finding used for in 3D vision?	Remember	The learner has to recall the concept active range finding used for in 3D vision .	CO5
8	Describe point-based representation in 3D vision.?	Understand	The learner has to recall the concept point-based representation .	CO5
9	What are volumetric representations in 3D vision?	Understand	The learner has to recall the concept of volumetric representations and .	CO5
10	How does 3D object recognition work?	Understand	The learner has to recall the concept 3D object recognition and .	CO5
11	What is 3D reconstruction in computer vision?	Understand	The learner has to recall the concept of 3D reconstruction and extend its uses.	CO5
12	What does the term "motion" refer to in 3D vision?	Remember	The learner has to recall the motion.	CO5
13	How does triangulation aid in 3D vision?	Remember	The learner has to recall the concept triangulation aid in 3D vision and outline its uses.	CO5
14	What is bundle adjustment in 3D vision?	Understand	The learner has to recall the concept of bundle adjustment in 3D vision and outline its applications	CO5
15	Explain translational alignment in 3D vision?	Understand	The learner has to recall the concept translational alignment .	CO5
16	How is parametric motion represented in 3D vision?	Understand	The learner has to define the concept of parametric motion represented in 3D vision .	CO5
17	Describe spline-based motion in 3D vision.?	Remember	The learner has to recall the concept of spline-based motion.	CO5
18	Define bundle adjustment in 3D vision?	Remember	The learner has to recall the concept bundle adjustment .	CO5

19	Describe point-based representation in 3D vision?	Understand	The learner has to recall the concept of point-based representation and explain its uses.	CO5
20	Define the shape from texture in 3D vision?	Understand	The learner has to recall the concept of shape from texture in 3D vision and outline its applications.	CO6
MODULE V				
APPLICATIONS				
PART A- PROBLEM SOLVING AND CRITICAL THINKING QUESTIONS)				
1	Analyze the psychological and emotional impact of revisiting photo albums, both physical and digital, on individuals and families. How do these albums contribute to a sense of identity and nostalgia? a) What role does reminiscence therapy, using photo albums as triggers for memory recall, play in improving cognitive well-being among the elderly? b) How do digital photo albums facilitate the sharing of family histories and traditions across generations?	Apply	The learner has to recall the concepts of the psychological and emotional impact of revisiting photo albums, both physical and digital, on individuals and families.	CO6

2.	Discuss the real-world applications and significance of face detection, including facial authentication, emotion recognition, and privacy protection. How does face detection enhance the capabilities of these systems? a) What role does reminiscence therapy, using photoalbums as triggers for memory recall, play in improving cognitive well-being among the elderly? b) What ethical considerations apply to face detection in contexts like public surveillance and biometric data storage?	Understand	The learner has to recall the concept of real-world applications and significance of face detection, including facial authentication, emotion recognition, and privacy protection.	CO6
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3	What are "Active Appearance Models (AAMs)" and "3D Shape Models of Faces," and how do they contribute to the field of computer vision, facial analysis, and facial recognition? a) How do AAMs and 3D Shape Models differ in their representation and capabilities for facial analysis? b) What are the historical developments and key milestones in the development of these models?	Apply	The learner has to recall the concept of Active Appearance Models (AAMs) and "3D Shape Models of Faces," and how do they contribute to the field of computer vision, facial analysis, and facial recognition	CO6
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4	Explore the trade-offs between model complexity and computational efficiency in foreground-background separation techniques. How do different methods balance these factors, and what are the implications for real-time applications? a) How can parallel processing and GPU acceleration be used to improve the speed of foreground-background separation algorithms? b) What are the considerations when selecting a separation method for real-time video processing or augmented reality applications?	Understand	The learner has to recall the concept of complexity and computational efficiency in foreground-background separation techniques.	CO6
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5	Consider the impact of foreground-background separation on the user experience and interactions in augmented reality (AR), virtual reality (VR), and mixed reality (MR). How do these techniques enhance immersive experiences and interactions with the digital and physical worlds? a) How does foreground-background separation contribute to realistic occlusion of virtual objects in AR and MR environments? b) What opportunities do emerging AR and VR applications offer for foreground-background separation in gaming, education, and training scenarios?	Understand	The learner has to recall the concept of foreground-background separation on the user experience and interactions in augmented reality (AR), virtual reality (VR), and mixed reality (MR) .	CO6
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6	What is "Separation" in the context of particle filters, and why is it a critical aspect in various fields, including target tracking, robotics, and state estimation? a) How does separation contribute to improving the accuracy and robustness of particle filter-based systems? b) What are the historical developments and key milestones in the study of separation in particle filters?	Understand	The learner has to recall the concept of context of particle filters, and why it is a critical aspect in various fields .	CO6
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7	Explore the trade-offs between model complexity and computational efficiency in Chamfer matching algorithms. How do different methods balance these factors, and what are the implications for real-time applications? a) How can optimization techniques, parallel processing, and hardware acceleration improve the speed of Chamfer matching algorithms? b) What are the considerations when selecting a Chamfer matching approach for resource-constrained devices like mobile robots?	Understand	The learner has to recall the concept of model complexity and computational efficiency in Chamfer matching algorithms.	CO6
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8	What is the significance of combining views from multiple cameras in computer vision, surveillance, and 3D reconstruction? How does this multi-view approach enhance the quality and richness of visual data analysis? a) How does combining multiple camera views improve object tracking and recognition accuracy in crowded or dynamic scenes? b) What are the historical developments and milestones in the field of multi-camera systems and vision?	Understand	The learner has to recall the concept of significance of combining views from multiple cameras in computer vision, surveillance, and 3D reconstruction.	CO6
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9	Discuss the real-world applications and benefit of combining views from multiple cameras in domains like security, autonomous vehicles, and augmented reality. How do these applications leverage the synergy of multiple perspectives? a) How are multi-camera setups used in surveillance systems for monitoring large areas and tracking suspicious activities? b) What role does multi-camera data fusion play in enhancing situational awareness and obstacle detection in autonomous vehicles?	Apply	The learner has to recall the concept of real-world applications and benefit of combining views from multiple cameras in domains like security.	CO6
10	Explore the trade-offs between the level of detail in gait analysis and the complexity of measurement techniques. How do different methods balance these factors, and what are the implications for practical applications??	Apply	The learner has to recall concept level of detailing gait analysis and the complexity of measurement techniques	CO6
PART-B LONG ANSWER QUESTIONS				
1	How does object detection and recognition contribute to photo album applications?	Understand	The learner has to recall the concept of photo album applications and outline how tools to manage and present their visual memories in an organized and user-friendly manner	CO6

2	What are photo album applications in the context of digital media, and why are they important?	Understand	The learner has to recall the concept photo album applications and extends its applications.	CO6
3	How is face detection utilized in human-computer interaction and entertainment applications?	Understand	The learner has recall the concept of face detection utilized in human-computer interaction and Extend its applications.	CO6
4	What are the key challenges faced in face detection?	Understand	The learner has to recall the concept of face detection and demonstrates face detection applied in security and surveillance systems	CO6
5	How can face recognition be used in healthcare and healthcare-related applications?	Understand	The learner has to recall the concept of face recognition and extend the its applications.	CO6
6	How do eigenfaces compare to more modern facial recognition techniques?	Understand	The learner has to Recall the concept eigenfaces and relate how it will work.	CO6
7	What are 3D Shape Models of faces and how do they differ from AAMs??	Understand	The learner has to recall the concept of the 3D Shape Models of faces and extends its applications.	CO6
8	What are some applications of Active Appearance Models (AAMs) in computer vision?	Understand	The learner has to recall the concept of Active Appearance Models and Extend its applications .	CO6
9	How have deep learning techniques revolutionized foreground-background separation?	Understand	The learner has to recall the concept of revolutionized foreground-background separation and outlines specific applications of foreground-background separation.	CO6
10	What is foreground-background separation in computer vision, and why is it important?	Understand	The learner has to recall the concept foreground-background separation and extends its applications	CO6

11	What are particle filters, and how do they work?	Understand	The learner has to recall the concept of particle filters and Outline particle filters contribute to environmental monitoring and weather forecasting.	CO6
12	What are some practical applications of chamfer matching?	Understand	The learner has to recall the concept of chamfer and extend its limitations of chamfer matching	CO6
13	What is object tracking, and how does it work?	Understand	The learner has recall the concept of object tracking to Extend its applications.	CO6
14	What are occlusions in the context of object tracking, and why are they challenging to handle?	Understand	The learner has to recall the concept of occlusions object tracking outline practical applications of object tracking with occlusion handling.	CO6
15	How do object tracking algorithms handle occlusions?	Understand	The learner has to recall the concept of object tracking algorithms summerise statistical anomaly detection and rule-based intrusion detection.	CO5
16	What is the process of combining views from multiple cameras?	Understand	The learner has to recall the concept of combining views from multiple cameras outline its uses.	CO6
17	What are the applications of combining views from multiple cameras?	Understand	The learner has to recall concept of combining views from multiple cameras Extend its applications.	CO6
18	What are some challenges in combining views from multiple cameras?	Understand	The learner has to recall the concept of combining views from multiple cameras	CO6
19	What are the primary applications of human gait analysis in healthcare and rehabilitation?	Understand	The learner has to recall the concept of human gait analysis in healthcare and rehabilitation .	CO6

20	How does gait analysis contribute to the field of ergonomics and human factors engineering?	Understand	The learner has to recall the concept gait analysis extend the its applications.	CO6
PART-C-SHORTANSWERQUESTIONS				
1	How is a photo album application useful in image processing?	Remember	The learner has to recall the concept of photo album .	CO6
2	What is face detection in image processing?	Remember	The learner has to recall the concept of face detection in image processing.	CO6
3	How does face recognition work?	Understand	The learner has to recall the concept of face recognition work to outline its applications.	CO6
4	Explain Eigen faces in face recognition?	Understand	The learner has to recall the concept of Eigen faces in face recognition .	CO6
5	What are active appearance and 3D shape models of faces?	Remember	The learner has to recall the concept active appearance and 3D shape models of faces .	CO6
6	How does foreground-background separation benefit image processing ?	Remember	The learner has to recall the concept foreground-background separation.	CO6
7	What are particle filters in image processing?	Understand	The learner has to recall the concept of particle filters in image processing.	CO6
8	Explain Chamfer matching in image processing?	Understand	The learner has to recall the concept of Chamfer matching in image processing .	CO6
9	How does tracking and occlusion handling work in image processing?	Understand	The learner has to recall the concept tracking and occlusion handling work extend its principles.	CO6
10	What are the advantages of combining views from multiple cameras ?	Remember	The learner has to recall the concept of combining views from multiple cameras.	CO6
11	What is human gait analysis used for?	Remember	The learner has to recall the concept of human gait analysis.	CO6

12	How does a photo album application help users manage their images?	Remember	The learner has to recall the concept of a photo album application .	CO6
13	How does face detection benefit security systems?	Remember	The learner has to recall the concepts of face detection benefit security systems.	CO6
14	How is face recognition applied in biometric authentication?	Understand	The learner has to recall concepts of the face recognition applied in biometric authentication.	CO6
15	Describe the process of Eigenfaces in face recognition.?	Understand	The learner has to recall the concept of Eigenfaces in face recognition .	CO6
16	What is the role of active appearance models in face analysis?	Remember	The learner has to recall the concepts of active appearance models in face analysis.	CO6
17	How does foreground-background separation assist in object segmentation?	Remember	The learner has to recall the concepts of foreground-background separation	CO6
18	Explain how particle filters improve tracking accuracy.?	Understand	The learner has to recall the concept of particle filters outline its applications.	CO6
19	What are the challenges in combining views from multiple cameras	Understand	The learner has to recall the concept of combining views from multiple cameras Extend its applications.	CO6
20	How does human gait analysis contribute to biometric identification?	Remember	The learner has to recall the concept of human gait analysis .	CO6

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